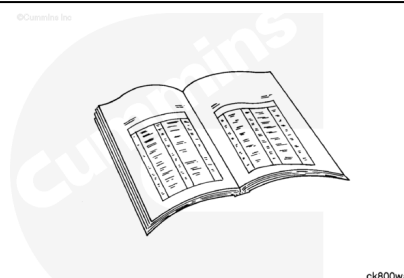


Trainings ▾

Preparatory Steps

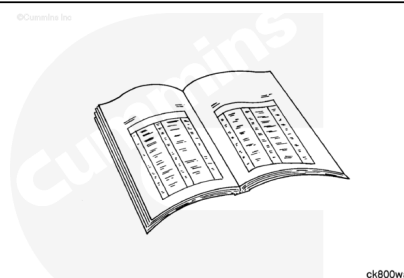
with Mechanically Actuated Injector

- Remove the Crankcase Breather. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html>).



with Electronically Actuated Injector

- Remove the fuel pump. Refer to Procedure 005-016 (</qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html>).
- Disconnect the crankcase breather. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html>).
- Remove the electronic control module (ECM) cooling plate. Refer to Procedure 008-128 (</qs3/pubsys2/xml/en/procedures/56/56-008-128.html>).
- Remove the pre-lubricating oil pump. Refer to Procedure 007-063 (</qs3/pubsys2/xml/en/procedures/56/56-007-063-tr.html>).
- Remove the water pump. Refer to Procedure 008-062 (</qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html>).
- Remove the starter motors. Refer to Procedure 013-020 (</qs3/pubsys2/xml/en/procedures/56/56-013-020-tr.html>).
- Remove the lubrication oil filters (Spin-On). Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html>).
- Remove the eliminator filter. Refer to Procedure 007-067 (</qs3/pubsys2/xml/en/procedures/56/56-007-067-tr.html>).



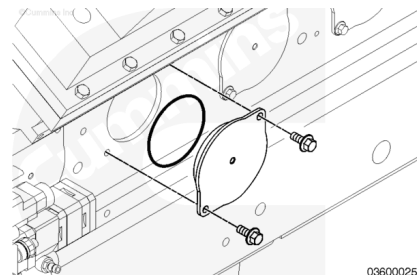
Remove

Standard

Remove the hand hole cover capscrews.

Remove the hand hole cover.

Remove and discard the o-ring.



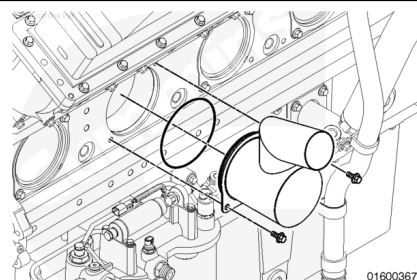
Block Mounted Crankcase Breather

Disconnect the crankcase breather pipe.

Remove the hand hole cover capscrews.

Remove the hand hole cover.

Remove and discard the o-ring.

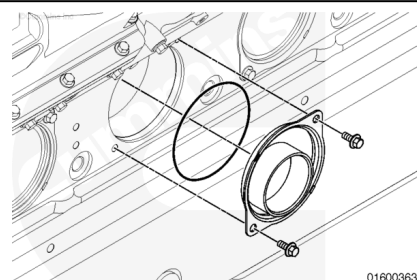


Eliminator™

Remove the hand hole cover capscrews.

Remove the Eliminator hand hole cover. Note the location where installed.

Remove and discard the o-ring.



Clean and Inspect for Reuse

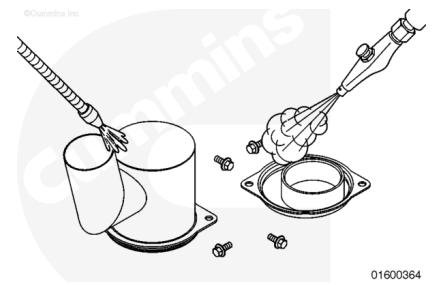
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

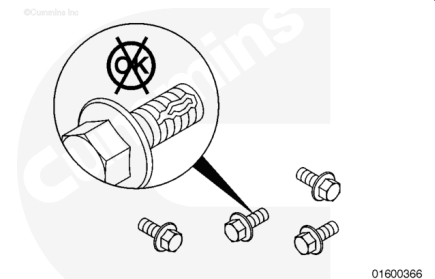


Use solvent to clean the hand hole covers and capscrews.
Dry with compressed air.

Inspect the hand hole covers for damage, holes, and cracks.
Replace if damaged.



Inspect the capscrews for damage and stripped threads.
Replace if damaged.



Install

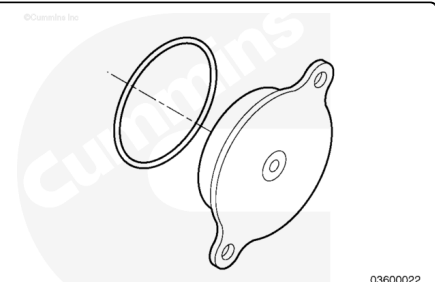
Standard

Note : : For Low NOx emissions capability products, the dipstick hand hole cover is reversed, with dipstick on the left, and filling point on the right. Ensure this is fitted, to prevent interference between the filling cap and Fuel Tubes.

Use the minimum amount of stretch and do **not** allow the o-ring to twist when installing in onto the hand hole cover.

Install a new o-ring into the groove of the hand hole cover.

Lubricate the o-ring seal with clean engine oil.



Do **not** pull the hand hole cover into the cylinder block by tightening the capscrews.

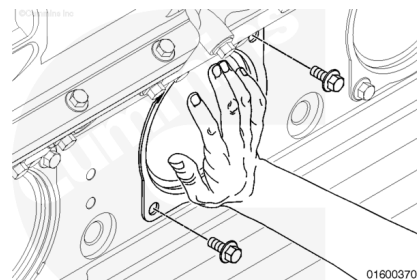
Place the hand hole cover into the cylinder block

Turn the hand hole cover until the location holes line up with the mounting holes on the block.

Apply firm pressure, pressing the hand hole cover into place.

Install and tighten capscrews.

Torque Value: 45 n•m [33 ft-lb]

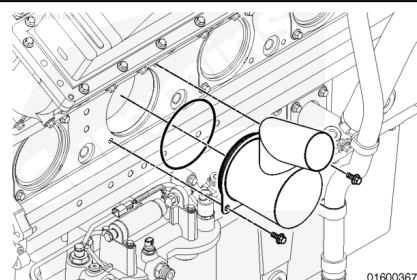


Block Mounted Crankcase Breather

Use the minimum amount of stretch and do **not** allow the o-ring to twist when installing in onto the hand hole cover.

Install a new o-ring into the groove of the hand hole cover.

Lubricate the o-ring seal with clean engine oil.



Do **not** pull the hand hole cover into the cylinder block by tightening the capscrews.

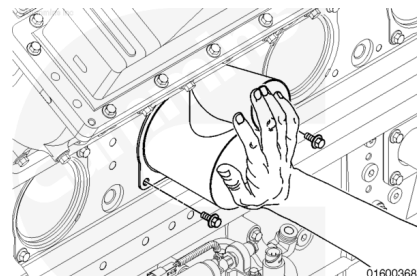
Place the hand hole cover into the cylinder block.

Turn the hand hole cover until the location holes line up with the mounting holes on the block.

Apply firm pressure, pressing the hand hole cover into place.

Install and tighten capscrews.

Torque Value: 45 n•m [33 ft-lb]

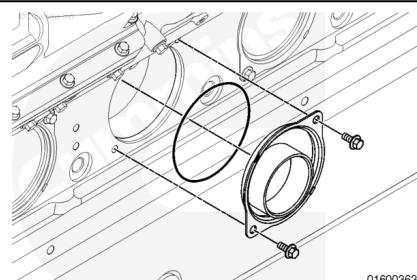


Eliminator™

Use the minimum amount of stretch and do **not** allow the o-ring to twist when installing in onto the hand hole cover.

Install a new o-ring into the groove of the hand hole cover.

Lubricate the o-ring seal with clean engine oil.



Do **not** pull the hand hole cover into the cylinder block by tightening the capscrews.

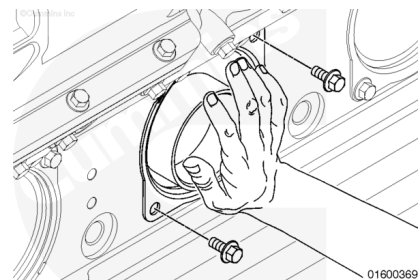
Place the hand hole cover into the cylinder block.

Turn the hand hole cover until the location holes line up with the mounting holes on the block.

Apply firm pressure, pressing the hand hole cover into place.

Install and tighten capscrews.

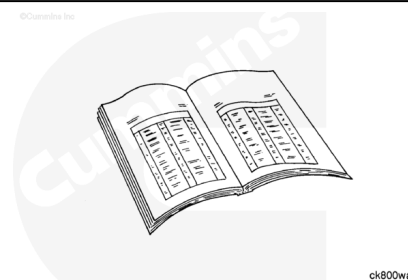
Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

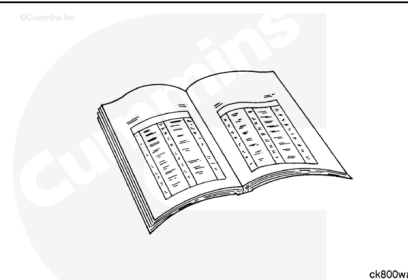
with Mechanically Actuated Injector

- Install the Crankcase Breather. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html>).



with Electronically Actuated Injector

- Install the eliminator filter. Refer to Procedure 007-067 (</qs3/pubsys2/xml/en/procedures/56/56-007-067-tr.html>).
- Install the lubrication oil filters (Spin-On). Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html>).
- Install the starter motors. Refer to Procedure 013-020 (</qs3/pubsys2/xml/en/procedures/56/56-013-020-tr.html>).
- Install the water pump. Refer to Procedure 008-062 (</qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html>).
- Install the pre-lubricating oil pump. Refer to Procedure 007-063 (</qs3/pubsys2/xml/en/procedures/56/56-007-063-tr.html>).
- Install the electronic control module (ECM) cooling plate. Refer to Procedure 008-128



(/qs3/pubsys2/xml/en/procedures/56/56-008-128.html).

- Connect the crankcase breather. Refer to Procedure 003-001 (/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html).
- Install the fuel pump. Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html).
- Operate the engine and check for leaks.

Last Modified: 12-Feb-2021
